

Azhar Ahmed

Data Scientist | ML Developer | IT Undergraduate

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Professional Summary

IT Undergraduate and certified Data Scientist with extensive project experience in Machine Learning, Deep Learning, and Computer Vision. Proficient in Python, PyTorch, TensorFlow, and modern Hugging Face transformers, with a proven track record of building, fine-tuning, and deploying high-accuracy models like Vision Transformers and CNNs. Eager to contribute strong analytical skills and production-ready coding practices to impactful AI development roles.

Technical Skills

Programming Languages: Python, Java, C++, SQL, JavaScript, HTML, CSS

Machine Learning: Scikit-learn, XGBoost, LightGBM, Regression, Classification, Clustering, Hyperparameter Tuning

Deep Learning: PyTorch, TensorFlow, Keras, ANN, CNN, RNN, LSTM, Vision Transformers (ViT), Transfer Learning

Computer Vision and NLP: OpenCV, YOLOv8, ResNet, VGG, Hugging Face Transformers, DistilBERT

Generative AI: LangChain, Retrieval-Augmented Generation (RAG), LLM APIs (Groq), Fine-Tuning

Data and Deployment: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, FastAPI, Flask, Docker

Projects

FasalGuard - Crop Disease Detection App

Technologies: Python, FastAPI, YOLOv8, ViT, React Native, Hugging Face, Groq

- Engineered a two-stage computer vision pipeline that detects and classifies crop diseases from a leaf photo across **38 disease classes and 13 crop species**.
- Fine-tuned a Vision Transformer on **87,000+ PlantVillage images** to reach **93% classification accuracy**.
- Delivered as a full-stack mobile app (React Native and FastAPI) with a Groq-powered chatbot for treatment advice, taking farmers from diagnosis to action.

PneumoDetect - Pneumonia Detection from Chest X-Rays

Technologies: Python, Flask, PyTorch, Vision Transformer (ViT), Docker

- Fine-tuned a Vision Transformer to detect pneumonia from chest X-rays, serving real-time predictions with confidence scores to support clinical decision-making.
- Optimized the model for speed and low memory use and containerized it with Docker for reliable deployment.

Source code: <https://github.com/azharahmedp/PneumoDetect>

SpamShield - Email Spam Detection with NLP

Technologies: Python, Flask, DistilBERT, Hugging Face Transformers, PyTorch

- Trained DistilBERT on **190,000+ emails** to catch subtle spam patterns that simpler filters miss.
- Deployed as a Flask web app with automatic GPU acceleration for real-time predictions.

Source code: <https://github.com/azharahmedp/SpamShield-BERT>

Heart Disease Prediction Using Machine Learning

Technologies: Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Built and compared multiple models to predict cardiovascular disease risk, tuning for recall to minimize missed diagnoses.

Source code: https://github.com/azharahmedp/Heart_Disease_Prediction_Using_Machine_Learning

California Housing Price Prediction Using ANN

Technologies: Python, TensorFlow, Keras, NumPy, Scikit-learn

- Developed a neural network with regularization to keep housing price predictions reliable on unseen data.

Source code: https://github.com/azharahmedp/California_Housing_Price_Prediction-ANN

Fashion MNIST Image Classification Using CNN

Technologies: Python, TensorFlow, Keras

- Built a convolutional neural network that classifies clothing images with **90%+ accuracy** across 10 categories.

Source code: https://github.com/azharahmedp/Fashion_MNIST-CNN

Education

Bachelor of Science in Information Technology, University of Sindh, Jamshoro

2025 - 2029 (Expected)

Certifications

Google AI Professional, Google (Coursera)

2026

Artificial Intelligence and Data Science, Saylani Mass IT Training

2026

Introduction to Modern AI, Cisco Networking Academy

2025

Certified Java Developer, People's IT Training Programme

2025

Certified Website Developer, People's IT Training Programme

2024